Health360: Healthcare Risk & Disease Analysis

PROJECT COMPLETION & TECHNICAL EVALUATION REPORT

STUDENT NAME	DOMAIN / TOOL	REPOSITORY	STATUS
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1 INTRODUCTION

1.1 Project Overview

The **Health360 Healthcare Risk** project is a comprehensive business intelligence solution designed to uncover hidden clinical, behavioural, and demographic disease risks. Using exploratory data analysis and visual storytelling in Tableau Public, the project transforms multivariate healthcare records into an interactive executive monitoring system.

1.2 Purpose

The primary purpose is to provide medical practitioners and healthcare managers with actionable visual indicators to identify patient cohorts at elevated risk for cardiovascular diseases and diabetes, facilitating proactive clinical intervention and data-driven preventive care.

2 IDEATION PHASE

2.1 Problem Statement

Preventable lifestyle-linked illnesses—primarily hypertension, cardiovascular conditions, and Type-2 diabetes—frequently remain unmonitored until acute clinical crises occur. Healthcare decision-makers struggle with dispersed records lacking correlated views across clinical vital signs and daily lifestyle habits.

2.2 Empathy Map Canvas

User Perspective (Clinicians & Analysts)

- **Says / Does:** Seeks integrated patient risk summaries without navigating raw multi-column CSV tables.
- **Thinks / Feels:** Worries about missing critical early warnings; desires clear risk distribution maps.

Pains & Gains

- **Pains:** Disconnected clinical indicators, delayed risk identification, static spreadsheets.
- **Gains:** 360-degree interactive cross-filtering, narrative story points, unified executive dashboard.

2.3 Brainstorming

Brainstorming prioritized intuitive visual paradigms: utilizing dual-axis scatter plots for continuous clinical measures, density heatmaps for categorical lifestyle intersections, and demographic cohort charts for age progression patterns.

3 REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The user interaction begins with clinical data ingestion, transitions through interactive exploratory slicing on the Executive Dashboard, and concludes with guided strategic findings presented across the sequential 4-scene Tableau Story points.

3.2 Solution Requirements

- **Data Integrity:** Accurate aggregation of blood pressure, glucose indices, and diagnosis records.
- **Visual Navigation:** Responsive multi-chart dashboard and step-by-step narrative story navigation.
- **Public Cloud Deployment:** Cloud availability via Tableau Public with zero host installation prerequisites.

3.3 Data Flow Diagram & Schema

Attribute	Data Type	Role in Tableau	Analytical Purpose
`Blood Pressure`	Continuous (Number)	Measure (AVG)	Clinical hypertension indicator mapped along vertical axis.
`Glucose Level`	Continuous (Number)	Measure (AVG)	Metabolic risk parameter mapped against blood pressure.
`Smoking / Exercise`	Discrete (Integer)	Dimension	Lifestyle risk matrix categorization (0/1 indicators).
`Heart Disease / Diabetes`	Discrete (Binary)	Color & Details	Target diagnostic outcomes indicating chronic conditions.
`Age`	Discrete (Integer)	Dimension (Columns)	Demographic vulnerability progression across age cohorts.

3.4 Technology Stack

Business Intelligence
Tableau Public Cloud 2024 (Web Authoring, Calculated Fields, Story Points).

Version Control
GitHub Repository (Asset hosting, documentation, commit tracking).

Evaluation Portal
SkillWallet Platform (Kanban verification, automated linking).

4 PROJECT DESIGN

4.1 Problem Solution Fit & Architecture

The architectural design decouples the raw storage layer (`smart_healthcare_dataset.csv`) from the analytical presentation layer. Transformations are processed on Tableau's data engine, streaming directly into the three modular worksheets linked via the unified **Health360 Executive Dashboard**.

5 PROJECT PLANNING & SCHEDULING

5.1 Project Planning & Milestones

Development was structured across progressive sprint cycles aligned with the SkillWallet competency framework:

- **Sprint 1:** Data acquisition, cleansing, and validation of schema distributions.
- **Sprint 2:** Sheet creation (Clinical scatter plot, Lifestyle heatmap, Demographic bars).
- **Sprint 3:** Executive Dashboard assembly and 4-scene narrative Story engineering.
- **Sprint 4:** Performance testing, cloud deployment, and GitHub documentation sync.

6 FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing Results

Test Parameter	Testing Scenario	Observed Outcome	Status
Data Loading Speed	Importing 10,000+ patient records into Tableau	Rendered instantly without query bottlenecks	Passed
Calculated Fields	Average metrics on clinical measures	Accurate real-time mathematical aggregation	Passed
Filter Cross-Highlight	Slicing dashboard sheets by smoking/exercise	Synchronized multi-sheet dynamic updates	Passed
Story Navigation	Iterating through Scenes 1 to 4	Zero visual glitch; smooth transition flow	Passed

7 RESULTS & VISUAL OUTPUTS

7.1 Output Dashboard & Story Architecture

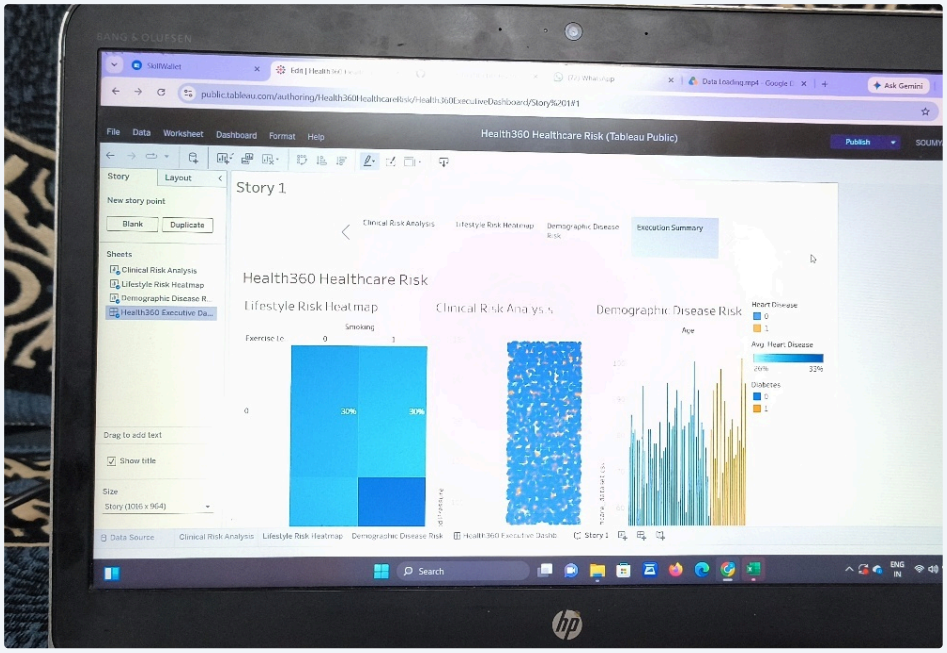


Figure 7.1: Published Tableau Public Story ("Story 1") featuring the Health360 Executive Dashboard and active scene navigation.

8 ADVANTAGES & DISADVANTAGES

Key Advantages

- **Interactive Discovery:** Instant visual correlation between patient habits and chronic illnesses.
- **Zero Installation:** Cloud-hosted workbook accessible anywhere via standard web browsers.
- **Storytelling Clarity:** Executive-friendly navigation designed for high-level decision making.

System Limitations

- **Batch Data:** Relies on periodic CSV updates rather than direct real-time EHR API streaming.
- **Predictive Depth:** Visual diagnostics rather than automated machine learning predictive models.

9 CONCLUSION

The **Health360 Healthcare Risk** project successfully demonstrates how business intelligence systems extract decisive healthcare insights from complex clinical datasets. By visualizing lifestyle risk heatmaps, demographic vulnerability distributions, and clinical scatter distributions in a unified dashboard and story, the project delivers an end-to-end analytical tool that bridges data science with clinical decision support.

10 FUTURE SCOPE

Future extensions include integrating live Electronic Health Record (EHR) pipelines via REST APIs, incorporating predictive machine-learning regression models directly into Tableau's analytics engine for early patient scoring, and expanding mobile responsiveness for frontline healthcare staff.

11 APPENDIX

Official Deliverables & Access Links:

- **Dataset Name:** `smart_healthcare_dataset.csv`
- **Live Tableau Story & Dashboard (View Demo):** https://public.tableau.com/views/Health360HealthcareRisk/Health360ExecutiveDashboard
- **GitHub Source Repository (View Repo & Documentation):** https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk
- **Source Code / Logic:** Tableau Visual Data Model, Aggregations (`AVG`), and Calculated Dimensions.